Filters in Servlet Programming

Kishan

Filters in Servlet Programming

- A Filter is a pre-processing component that executes before a Servlet program.
- Filters are configured with the same URL pattern as the associated Servlet, ensuring that both Filter and Servlet respond to the same requests.
- The Filter program has a higher priority in execution than the Servlet program, meaning it intercepts and processes requests before the Servlet.
- In real-time applications, Filters often contain security-related code, making the Filter a security layer for the Servlet program.

We use the following components to construct Filters:
1.Filter
2.FilterChain
3.FilterConfig

Kishan

Filter :

The Filter interface, part of the javax.servlet package, is implemented to create Filter programs.

public void init(FilterConfig config) throws ServletException;

Initializes the Filter when the server starts or when the Filter is first accessed. It uses a FilterConfig object to retrieve configuration information

public void doFilter(ServletRequest request, ServletResponse response, FilterChain chain) throws IOException, ServletException;

This is the core method of a Filter, called each time a request or response passes through the Filter. It allows for preprocessing of the request and post-processing of the response. The FilterChain object enables passing the request and response to the next entity in the chain (another Filter or the targeted Servlet)

public void destroy();

This method is called when the Filter is removed or the server shuts down, allowing for cleanup activities.

FilterChain :

FilterChain is an interface in the javax.servlet package, instantiated automatically during doFilter() method execution. It provides a mechanism to link multiple Filters and ultimately direct requests to the Servlet program.

Kishan

public void doFilter(ServletRequest request, ServletResponse response) throws IOException, ServletException;

This method is used within the doFilter() of a Filter to pass control to the next component in the chain, allowing the request to proceed through additional Filters or reach the final Servlet.

FilterConfig :

FilterConfig is an interface from javax.servlet package and which is instantiated automatically when filter-program is loaded for execution. This FilterConfig Object is loaded with Filter_name. To access FilterConfig object reference we must use init()-method. We need to use **<init-param>** which is a subtag of **<filter>** tag to initialize parameter-values to FilterConfig Object.

```
<web-app>
   ...
   <filter>
      <filter-name> FilterName </filter-name>
      <filter-class> FullyQualifiedFilterClassName </filter-class>
      <init-param>
         <param-name> paramName </param-name>
         <param-value> paramValue </param-value>
      </init-param>
   </filter>
   <filter-mapping>
      <filter-name> FilterName </filter-name>
      <url-pattern> /specificURLPattern </url-pattern>
   </filter-mapping>
   ...
</web-app>
```

1) The FilterConfig object is only accessible in the **init()** method of the filter.

2) The <filter> tag specifies the filter name and class, while the <init-param> subtag allows you to add specific configuration parameters.

3) The <filter-mapping> tag associates the filter with a URL pattern, specifying which requests the filter will intercept

Kishan